THE NATIONAL INSTITUTE OF ENGINEERING

Mysuru – 570 008

VIRTUAL LAB REPORT

COURSE: Linear Integrated Circuits
COURSE CODE: BEC405D

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ACADEMIC YEAR 2025-2026

IIT ROORKEE · VIRTUAL LABS
Analog & Digital Electronics II

Characteristics and Parameters of the

Operational Amplifier IC-741

A Detailed Introduction to the Experiment

1. WHAT IS THIS EXPERIMENT ABOUT?

This experiment is conducted as part of the **Analog & Digital Electronics II** virtual laboratory hosted by **IIT Roorkee** on the National Virtual Labs (VLabs) platform. It is designed to give students of Electronics and Communication Engineering a thorough, hands-on understanding of the **real-world electrical behaviour** of one of the most fundamental components in analog circuit design — the **Operational Amplifier (Op-Amp)**, specifically the industry-standard **IC-741**.

The IC-741 is a general-purpose operational amplifier that has been in production for over five decades and continues to serve as the **reference device for understanding Op-Amp theory**. Despite the emergence of more advanced, higher-speed Op-Amps in modern applications, the 741 remains the pedagogical standard in undergraduate ECE curricula worldwide — precisely because its behaviour clearly illustrates both the **capabilities and the limitations** of a practical Op-Amp relative to the ideal theoretical model.

The central purpose of this experiment is to **bridge the gap between theory and reality**. In classroom lectures, the Op-Amp is treated as an ideal device — infinite gain, infinite bandwidth, zero offset. In practice, the IC-741 deviates from every one of these assumptions. Through five carefully structured sub-experiments, students directly measure these deviations, compute the governing parameters from first principles, and compare them against the **manufacturer's datasheet specifications**.

PLATFORM & CONTEXT

This experiment is part of the IIT Roorkee VLab series: 'Analog and Digital Electronics II'. The virtual simulation replicates a real breadboard setup — including a ±15 V DC power supply, IC-741 Op-Amp, precision resistors, capacitors, a Digital Multimeter (DMM), a CRO, and an AF Oscillator — allowing students to perform measurements without physical lab access while maintaining experimental rigour.

2. THE OPERATIONAL AMPLIFIER — BACKGROUND

2.1 Origin and Definition

An **Operational Amplifier** is a **direct-coupled, high-gain differential amplifier** integrated into a single IC package. The name 'operational' derives from its original use in analog computers of the 1940s and 1950s, where it was employed to perform mathematical **operations** — addition, subtraction, integration, and differentiation — on continuous electrical signals. With the addition of suitable external feedback networks, the modern Op-Amp serves an extraordinarily wide range of applications:

- **DC and AC signal amplification** — from millivolt sensor outputs to audio signals
- **Active filters** — low-pass, high-pass, band-pass, and notch configurations
- **Oscillators and waveform generators** — sine, square, triangle wave production
- **Comparators and Schmitt triggers** — threshold detection and signal conditioning
- **Voltage regulators and references** — precision DC supply generation
- **Instrumentation amplifiers** — high-accuracy differential measurement in noisy environments

2.2 Internal Architecture of the IC-741

The IC-741 is internally composed of a **three-stage cascade**, each stage serving a distinct and essential role in producing the overall Op-Amp behaviour:

Stage	Circuit Topology	Role & Significance
Input Stage	Dual-input, balanced-output differential amplifier	Provides the bulk of open-loop voltage gain. Establishes the high input impedance of the Op-Amp. The two differential inputs (+) and (−) are defined here.
Intermediate Stage	Dual-input, single-ended (unbalanced) output differential amplifier	Adds further voltage gain. Since direct coupling is used, the DC voltage at this stage's output is significantly above ground — a level translator

		(shifting) circuit corrects this, shifting the output back toward 0 V.
Output Stage	Push-pull complementary-symmetry pair	Boosts current drive capability so the Op-Amp can source/sink significant load current. Provides low output impedance for efficient power transfer. Increases the output voltage swing.

2.3 Schematic Symbol and Input Conventions

The Op-Amp schematic symbol is a triangle with two differential inputs and one output. The **non-inverting input (+)** at Pin 3 produces an output that is **in-phase** with the applied signal. The **inverting input (–)** at Pin 2 produces an output that is **180° out of phase** (opposite polarity). Supply voltages (+15 V and −15 V) are connected to Pins 7 and 4 respectively. The processed output appears at Pin 6.

2.4 The Ideal Op-Amp — Theoretical Benchmark

Circuit analysis of Op-Amp circuits is greatly simplified by assuming ideal behaviour. The following table defines the ideal characteristics — each of which this experiment deliberately tests and measures the deviation from:

Ideal Parameter	Ideal Value	Why It Matters in Circuit Design
Voltage Gain (A)	Infinite (∞)	Allows precise closed-loop gain to be set entirely by external resistors, independent of the Op-Amp itself.
Input Resistance (Ri)	Infinite (∞)	Draws zero current from the signal source — no loading effect on the preceding stage.
Output Resistance (Ro)	Zero (0 Ω)	Output voltage is unaffected by load; the Op-Amp can drive any impedance without loss.
Bandwidth	Infinite (0 to ∞ Hz)	Amplifies all frequencies equally — no frequency-dependent gain roll-off.
CMRR	Infinite (∞ dB)	Perfectly rejects common-mode noise (e.g., power-line hum present on both inputs simultaneously).
Slew Rate	Infinite	Output voltage can change instantaneously — no lag when tracking fast input signals.
Input Offset Voltage	Zero (0 V)	Output is exactly zero when both inputs are at the same potential.

KEY INSIGHT — IDEAL VS. PRACTICAL

No real Op-Amp achieves any of these ideal values. The IC-741 has a finite open-loop gain (~200,000 V/V), a finite input resistance (~2 MΩ), a non-zero output resistance (~75 Ω), a limited bandwidth (~1 MHz), a finite CMRR (~90 dB), and a critically slow slew rate of only 0.5 V/µs. This experiment measures these deviations directly, giving students a quantitative appreciation of where and why ideal analysis breaks down in practice.

3. PARAMETERS UNDER STUDY — WHAT, WHY, AND HOW

The experiment is structured into five sequential sub-experiments, each targeting a specific Op-Amp parameter. The order is deliberate — each result feeds into the calculation of the next. The five parameters, their physical meaning, measurement circuits, and governing formulae are detailed below.

3.1 Input Bias Current (IB)

The IC-741's input stage consists of two NPN transistors in a differential configuration. Like all BJTs, these transistors require a small **DC base current** to remain in their active region of operation. This current — called the **Input Bias Current** — flows continuously into (or out of) the input terminals even when no signal is intentionally applied. If this current passes through feedback or source resistors, it produces an unwanted **voltage drop**, which appears as an erroneous output offset. In precision DC amplifiers and integrators, failing to account for bias current leads to significant measurement errors.

The bias current at each terminal is measured individually. The **inverting bias current IB2** is measured by connecting Rf = 470 kΩ between Pin 2 and Pin 6 (output), grounding Pin 3, and reading the output voltage Vo on the DMM. Similarly, **IB1** is measured with R1 = 470 kΩ at Pin 3, with Pin 2 connected to Pin 6 via Rf. The average of both gives IB:

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IB2 = Vo / Rf
IB1 = Vo / R1
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IB = (IB1 + IB2) / 2
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IC-741C Specification: IB(max) = 500 nA | Precision 741C: ±7 nA

3.2 Input Offset Current (Iio)

Even within a single IC chip, no two transistors are fabricated with perfectly identical characteristics. As a result, **IB1 and IB2 are not equal**. The **Input Offset Current** is defined as the magnitude of their difference, and directly quantifies the degree of transistor mismatch in the input stage:

$$Iio = |IB1 - IB2| \rightarrow \text{measured as: } Iio = Vo / Rf$$

For this measurement, both R1 and Rf are connected simultaneously (to Pins 3 and 2→6 respectively), and the resulting Vo reflects the combined effect of both bias currents. The smaller the Iio, the better matched the input transistors — and the lower the drift in precision applications. **IC-741C spec: Iio(max) = 200 nA.**

3.3 Input Offset Voltage (Vio)

Even with both inputs shorted together to the same potential, the internal asymmetry of the Op-Amp causes the output to sit at a non-zero voltage. The **Input Offset Voltage** is the small differential voltage that must be applied between the two input pins to force the output to exactly **zero volts**. It is a direct measure of DC imperfection in the input stage.

$$Vio = (Vo - Iio \times Rf) / (1 + Rf / Ri)$$

The circuit uses Ri = 100 Ω and Rf = 10 kΩ, giving a closed-loop gain of (1 + Rf/Ri) = 101. This large gain amplifies the offset, making it easily measurable on a standard DMM. The term (Iio × Rf) corrects for the contribution of offset current to Vo. **IC-741C spec: Vio(max) = 6 mV DC.**

3.4 Slew Rate (SR)

The **Slew Rate** is arguably the most practically important parameter studied in this experiment. It defines the **maximum rate at which the output voltage can change**, expressed in V/µs. It is physically limited by the rate at which the internal compensation capacitor (≈30 pF in the 741) can be charged or discharged by the input differential pair's tail current.

When an input signal demands an output voltage change **faster** than the slew rate permits, the Op-Amp cannot keep up. Instead of following the input waveform faithfully, the output charges at its maximum rate — producing a **triangular wave** (constant slope = SR). This condition — called **slew-rate limiting** — is exactly what is observed on the CRO in Sub-experiment D. The frequency at which the output just transitions into a triangular waveform is recorded as **fmax**, and SR is computed as:

$$SR = 2\pi \times fmax \times Vm \times 10^{-6} \quad [V/\mu s] \qquad (Vm = 3\ V,\ fixed)$$

The **IC-741C slew rate of ≈0.5 V/µs** is one of its most cited limitations. This limits undistorted full-power output to frequencies below ≈26.5 kHz (for a ±15 V swing). In contrast, pin-compatible modern replacements — the LF351 (13 V/µs) and LM318 (70 V/µs) — handle frequencies in the MHz range.

Understanding this limitation is fundamental to selecting the correct Op-Amp for high-frequency applications such as active filters, oscillators, and wideband amplifiers.

3.5 Common-Mode Rejection Ratio (CMRR) — Contextual Overview

While CMRR is not directly measured in this virtual simulation, it is an essential parameter that defines the Op-Amp's **noise immunity**. CMRR is the ratio of the differential voltage gain (Ad) to the common-mode voltage gain (Acm):

```
CMRR  =  Ad / Acm          [expressed in dB = 20 log(CMRR)]
```

A high CMRR means that signals appearing equally on both inputs — such as 50 Hz mains interference inductively coupled to both wires of a sensor lead — are heavily attenuated at the output, while the actual differential signal of interest is faithfully amplified. **IC-741C: CMRR = 90 dB (typical) under Rs < 10 kΩ.** Precision 741C: 120 dB.

4. THE VIRTUAL SIMULATION — SETUP & OBSERVATIONS

The following screenshots document each sub-experiment as performed in the **IIT Roorkee Virtual Lab simulator**. Each screenshot captures the final verified circuit connection on the virtual breadboard and the corresponding DMM or CRO reading, along with the computed result displayed by the simulator.

Sub-Experiment A1 — Measurement of Inverting Bias Current (IB2)

Circuit configuration: Rf = 470 kΩ connected between Pin 2 (inverting input) and Pin 6 (output). Pin 3 (non-inverting input) is grounded. ±15 V supply applied to Pins 7 and 4. The DMM reads the output voltage Vo, from which IB2 is derived.

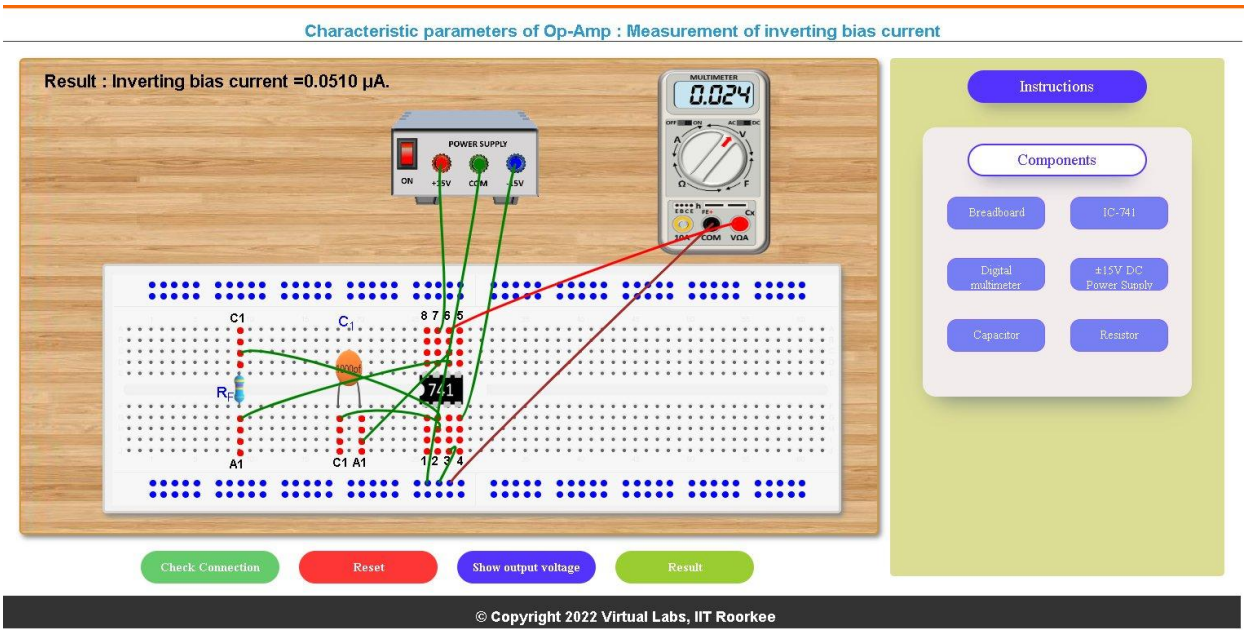

Fig. 1 — Virtual simulation: Inverting bias current measurement. DMM reads Vo = 0.024 V. Result: IB2 = 0.0510 µA (51.0 nA)

Sub-Experiment A2 — Measurement of Non-Inverting Bias Current (IB1)

Circuit configuration: R1 = 470 kΩ connected to Pin 3 (non-inverting input). Rf = 470 kΩ connects Pin 2 to Pin 6. The DMM reads Vo from which IB1 is computed. Note the negative polarity of the DMM reading — indicating the non-inverting bias current flows in the opposite direction relative to the output.

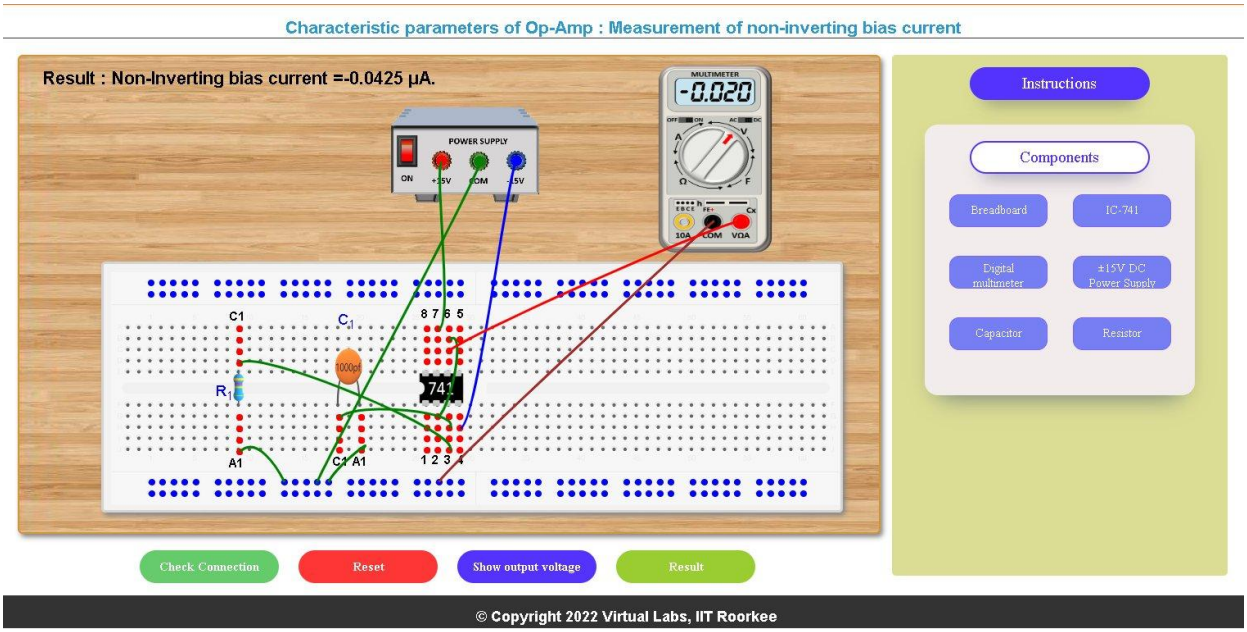

Fig. 2 — Virtual simulation: Non-inverting bias current measurement. DMM reads Vo = −0.020 V. Result: IB1 = −0.0425 µA (−42.5 nA)

Sub-Experiment B — Measurement of Input Offset Current (Iio)

Both R1 and Rf (each 470 kΩ) are active simultaneously — R1 at Pin 3, Rf from Pin 2 to Pin 6. Two capacitors (C1, C2) provide supply bypassing. The resulting Vo reflects the net effect of both bias currents through their respective resistors.

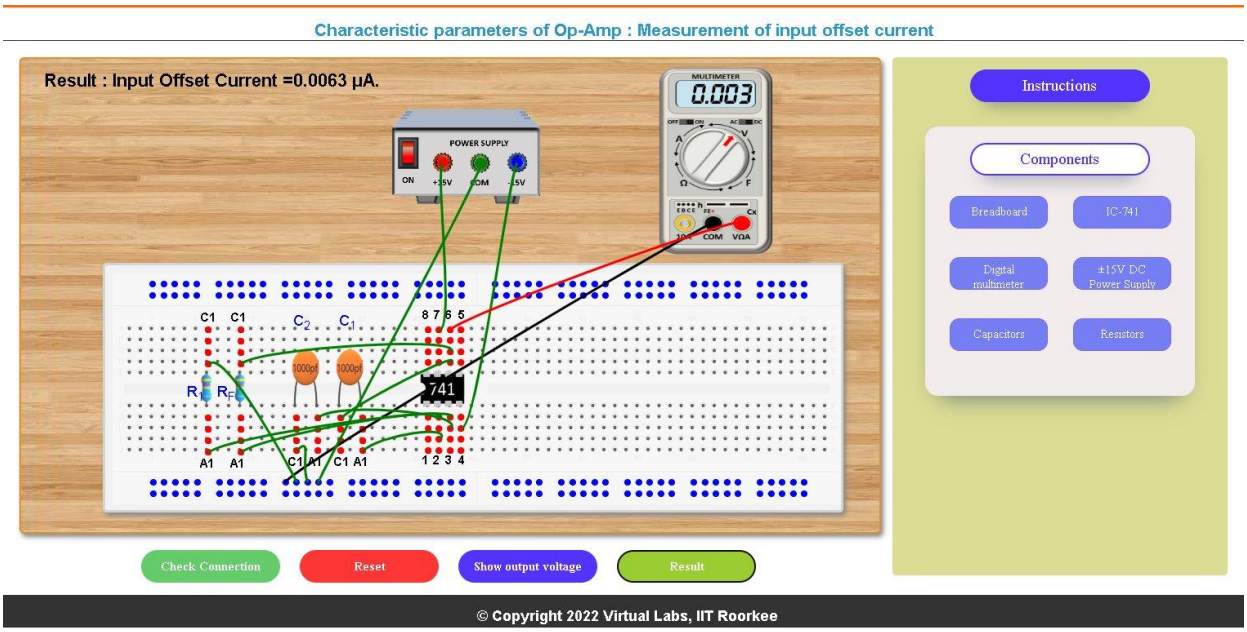

Fig. 3 — Virtual simulation: Input offset current measurement. DMM reads Vo = 0.003 V. Result: Iio = 0.0063 µA (6.3 nA)

Sub-Experiment C — Measurement of Input Offset Voltage (Vio)

The closed-loop inverting configuration uses Ri = 100 Ω and Rf = 10 kΩ, giving a closed-loop gain of 101. This magnifies the offset voltage by a factor of 101, making it measurable on the DMM. Three capacitors are used for supply decoupling. The formula accounts for offset current contribution.

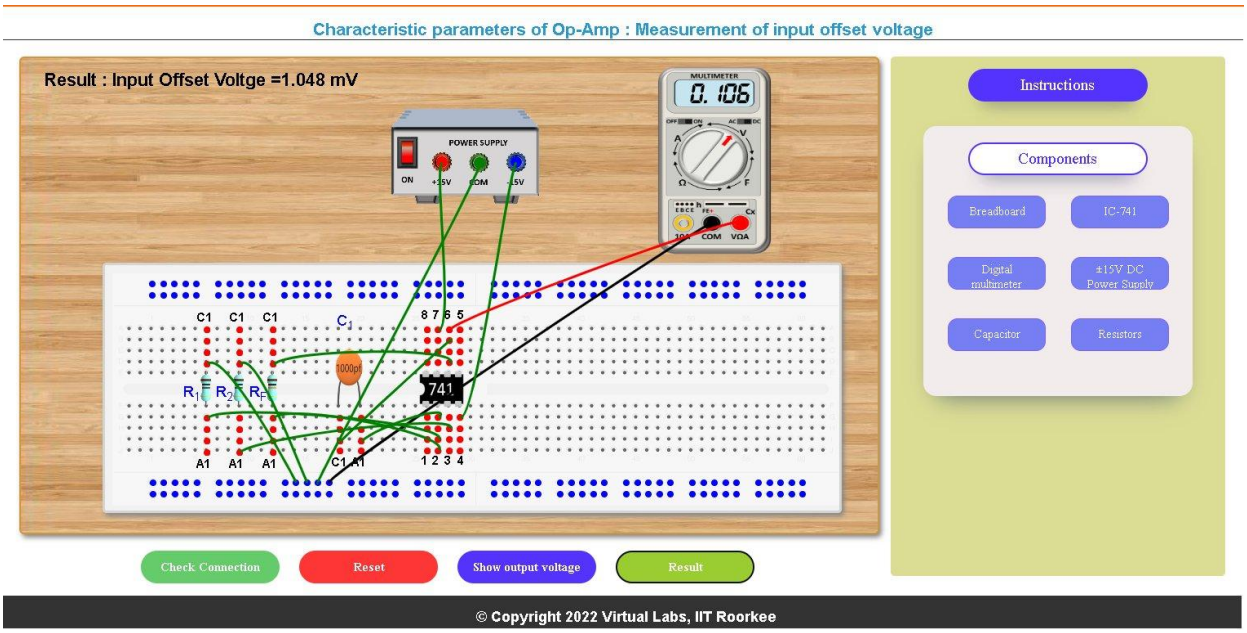

Fig. 4 — Virtual simulation: Input offset voltage measurement. DMM reads Vo = 0.106 V. Result: Vio = 1.048 mV

Sub-Experiment D — Measurement of Slew Rate (SR)

The Op-Amp is configured as a unity-gain voltage follower (Pin 6 directly to Pin 2). A square/sine wave of Vm = 3 V is applied from the AF Oscillator to Pin 3. The frequency is increased until the CRO output (CH2) transitions into a triangular wave. Both waveforms — sinusoidal input on CH1 (top trace) and triangular slew-rate-limited output on CH2 (bottom trace) — are clearly visible.

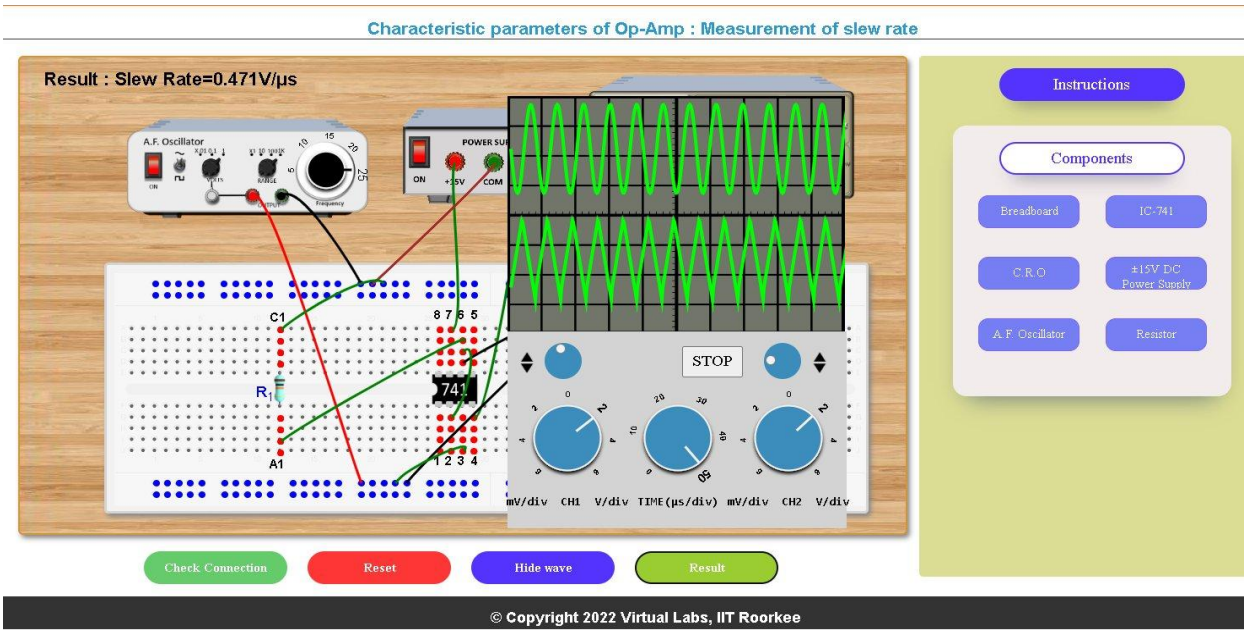

Fig. 5 — Virtual simulation: Slew rate measurement. CRO shows sinusoidal input (CH1, top) and triangular slew-limited output (CH2, bottom). Result: SR = 0.471 V/µs

5. SIGNIFICANCE —

Understanding Op-Amp parameters is not a purely academic exercise. These parameters determine whether a circuit works reliably in the real world or fails in ways that are difficult to diagnose. Consider the following practical consequences:

- **Precision instrumentation** — A thermocouple amplifier with 100× gain and an Op-Amp with Vio = 1 mV will report a 100 mV DC error — equivalent to a 2.5°C temperature error for a typical K-type thermocouple. Offset null compensation is essential.
- **Integrator circuits** — An ideal integrator accumulates the integral of Vio indefinitely, causing the output to ramp to the supply rail. In practice, integrators require bias-current compensation resistors and offset null trimming to remain stable.
- **Audio and communication amplifiers** — A 741 Op-Amp used in an audio filter beyond 20 kHz will produce slew-rate distortion on transient signals, degrading high-frequency fidelity — this is why audio-grade Op-Amps (TL072, NE5532) specify slew rates of 13–20 V/µs or higher.
- **Medical instrumentation (ECG/EEG)** — Differential biosignal amplifiers require extremely high CMRR (>100 dB) to reject 50/60 Hz mains interference that appears equally on both body electrodes.
- **Op-Amp selection** — Engineers routinely compare slew rate, offset voltage, CMRR, and bias current when selecting an Op-Amp from a datasheet. This experiment gives students the measurement vocabulary and physical intuition to interpret those specifications meaningfully.

SUMMARY OF ALL PARAMETERS MEASURED IN THIS EXPERIMENT

① IB2 — Inverting Bias Current: 51.0 nA (spec ≤ 500 nA ✓) ② IB1 — Non-Inverting Bias Current: −42.5 nA (spec ≤ 500 nA ✓) ③ IB — Average Input Bias Current: 46.75 nA (spec ≤ 500 nA ✓) ④ Iio — Input Offset Current: 6.3 nA (spec ≤ 200 nA ✓) ⑤ Vio — Input Offset Voltage: 1.048 mV (spec ≤ 6 mV ✓) ⑥ SR — Slew Rate: 0.471 V/µs (spec ≈ 0.5 V/µs ✓) All experimentally measured values are well within the IC-741C manufacturer's specified limits, validating both the measurement methodology and the fidelity of the IIT Roorkee virtual simulation environment.